6-6a Valuation at Lower of Cost or Market

If the market value is lower than the purchase cost, the **lower-of-cost-or-market (LCM) method** (A method of valuing inventory that reports the inventory at the lower of its cost or current market value (net realizable value).) is used to value the inventory. *Market,* as used in *lower of cost or market*, is the **net realizable value** (The estimated selling price of an item of inventory less any direct costs of disposal, such as sales commissions; the value of the receivables reduced to the amount that is expected to be collected or realized, computed as accounts receivable less allowance for doubtful accounts.) of the inventory. Net realizable value is determined as follows:

$$\text{Net Realizable Value} = \text{Estimated Selling Price} - \text{Direct Costs of Disposal}$$

Direct costs of disposal include selling expenses such as special advertising or sales commissions.

To illustrate, assume the following data about an item of damaged inventory:

Original cost	$1,000
Estimated selling price	800
Estimated selling expenses	150

In applying LCM, the market value of the inventory is $650, computed as follows:

$$\text{Market Value (Net Realizable Value)} = \$800 - \$150 = \$650$$

Thus, the inventory would be valued at $650, which is the lower of its cost of $1,000 and its market value of $650.

The lower-of-cost-or-market method can be applied in one of three ways. The cost, market price, and any declines could be determined for the following:

1. Each item in the inventory
2. Each major class or category of inventory
3. Total inventory as a whole

The amount of any price decline is included in the cost of goods sold. This, in turn, reduces gross profit and net income in the period in which the price declines occur. This matching of price declines to the period in which they occur is the primary advantage of using the lower-of-cost-or-market method.

Link to Best Buy

Best Buy values its inventory at lower of cost or market based upon cost and the amount it expects to realize from the sale.

Ethics in Action

Where's the Bonus?

Managers are often given bonuses based on reported earnings numbers. This can create a conflict. For example, LIFO can improve the value of the company through lower taxes. However, using LIFO also lowers management bonuses that are based on reported income. Thus, a manager might use FIFO to maximize his or her bonus even though LIFO is better for the company.

To illustrate, assume the following data for 400 identical units of Item A in inventory on December 31:

Cost per unit	$10.25
Market value (net realizable value) per unit	9.50

Since the market value of Item A is $9.50 per unit, $9.50 is used under the lower-of-cost-or-market method.

Exhibit 10 illustrates applying the lower-of-cost-or-market method to (1) each inventory item (Echo, Foxtrot, Sierra, Tango), (2) each major class of inventory (Class 1, Class 2), and (3) inventory in total. As applied on an item-by-item basis, the total lower-of-cost-or-market is $15,070, which is a market decline of $450 ($15,520 − $15,070). As applied to each class of inventory, the inventory would be valued at $15,412, which is a market decline of $108 ($15,520 − $15,412). Finally, if the lower-of-cost-or-market method is applied to the total inventory, the inventory would be valued at $15,472, which is a market decline of $48 ($15,520 − $15,472). The market declines under each approach ($450, $108, or $48) would be included in cost of goods sold.

Exhibit 10

Determining Inventory at Lower of Cost or Market (LCM)

Item	Inventory Quantity	Cost per Unit	Market Value per Unit (Net Realizable Value)	Cost	Market	Lower of Cost or Market (LCM)	
Echo	400	$10.25	$ 9.50	$ 4,100	$ 3,800	$ 3,800	(1) Applied Individual Item by Item
Foxtrot	120	22.50	24.10	2,700	2,892	2,700	
Sierra	600	8.00	7.75	4,800	4,650	4,650	
Tango	280	14.00	14.75	3,920	4,130	3,920	
Total				$15,520	$15,472	$15,070	
Class 1:							(2) Applied Class by Class
Echo	400	$10.25	$ 9.50	$ 4,100	$ 3,800		
Foxtrot	120	22.50	24.10	2,700	2,892		
Subtotal				$ 6,800	$ 6,692	$ 6,692	
Class 2:							
Sierra	600	8.00	7.75	$ 4,800	$ 4,650		
Tango	280	14.00	14.75	3,920	4,130		
Subtotal				$ 8,720	$ 8,780	8,720	
Total				$15,520	$15,472	$15,412	
Echo	400	$10.25	$ 9.50	$ 4,100	$ 3,800		(3) Applied in Total
Foxtrot	120	22.50	24.10	2,700	2,892		
Sierra	600	8.00	7.75	4,800	4,650		
Tango	280	14.00	14.75	3,920	4,130		
Total				$15,520	$15,472	$15,472	

Applying the lower-of-cost-or-market method on an item-by-item basis always gives the lowest value for inventory. Conversely, applying the lower-of-cost-or-market method to the total inventory always gives the highest value for inventory.

Link to Best Buy

The excess of cost over the amount **Best Buy** expects to receive from the sale of an item is called a markdown.

Check up Corner 6-4

Lower of Cost or Market

JJ's Electronics Company has three products in inventory (PCs, tablets, and smartphones). Each product's quantity, cost per unit, and market value per unit are as follows:

Item	Inventory Quantity	Cost per Unit	Market Value per Unit (Net Realizable Value)
PC	10	$175	$168
Tablet	12	132	150
Smartphone	8	199	187

Apply the lower-of-cost-or-market method to each inventory item in a form similar to Exhibit 10.